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Creating Virtual Height Loudspeakers for Dolby Atmos and Auro-3D Using VHAP

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ABSTRACT

This engineering brief evaluates the effectiveness of VHAP (virtual hemispherical amplitude panning) as a tool to create virtual height loudspeakers for Dolby Atmos 5.1.4 and Auro-3D 9.1 content. Previous studies showed that VHAP can effectively render elevated phantom images using only three or four ear-level loudspeakers. This study subjectively compared the 4-channel and 3-channel VHAP stimuli rendered against the original 5.1.4, in terms of sense of height, sense of width and overall quality of experience. Five excerpts from Dolby Atmos and Auro-3D content of different types (film, classical and EDM) were tested. Results suggest that VHAP is able to create virtual height loudspeakers with no significant level of degradation, and for certain stimuli it can be slightly preferred to the original

1 Introduction

Three-dimensional (3D) audio formats such as Dolby Atmos, Auro-3D and 22.2 employ the so-called ‘height’ channels to provide the user with a more realistic and immersive spatial experience. The Dolby Atmos 5.1.4 and Auro-3D 9.1 systems are being widely adopted in audio-visual receivers (AVR) for home-cinema. Despite their technical differences, they both use the conventional 5.1 layout for the ear-height loudspeakers and require four height loudspeakers elevated at around 30° to 45° from the listening position. However, there might be practical constraints in installing such elevated loudspeakers in home environments. Some recent sound bar systems are equipped with “up-firing” and “side-firing” drivers to exploit reflections from the ceiling and side walls as an alternative to the real height and surround loudspeaker signals.

Although such soundbars certainly provide a convenient solution for enjoying 3D content in homes, the experience might depend on the acoustics of the room and the quality of the drivers used. Also, for those who already have AVR and 5.1

loudspeaker systems, it might be desired to have a way to create the height images virtually using the existing system. This could be realised using the VHAP (Virtual Hemispherical Amplitude Panning) method [1], which was developed based on research conducted on the phantom image elevation effect [2]. A phantom centre image produced between two loudspeakers would be elevated, and its degree tends to increase as the loudspeaker base angle increases (perceived elevation is roughly twice as high as the base angle, e.g. directly overhead with loudspeakers at the sides). Based on this principle, VHAP attempts to position a phantom image at an arbitrary position over the upper-hemisphere using a constant-power panning between four ear-level loudspeakers placed at the listener’s left side, right side, front centre and back centre, as illustrated in Fig. 1. More details about the algorithm can be found in [1].

The aim of the present study was to evaluate the effectiveness of VHAP as a tool to create virtual height loudspeakers for Dolby Atmos and Auro-3D content. In addition, it was of interest to examine if there would be any perceived difference between the

original 4-channel VHAP and a 3-channel compromised version without the back loudspeaker.

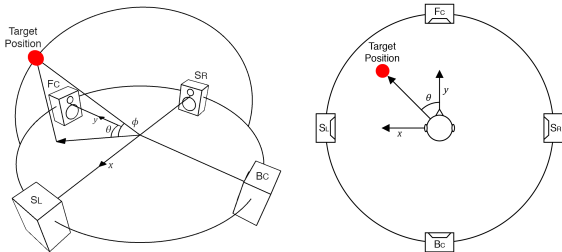


Fig. 1. Aerial and top views of loudspeaker arrangement for VHAP [1].

2 Method

2.1 Stimuli Creation

Excerpts from practical Dolby Atmos 5.1.4 and Auro-3D 9.1 samples of different genres (film, classical, environmental and electronic) were chosen for the listening test, consisting of the following:

- Aeroplane: a Boeing 747 aeroplane taking off, taken from Dolby Atmos demo disc 2017.
- Birds: an animation soundtrack with a flying bird from one side to another, taken from Auro-3D demo disc vol. 2.
- EDM: an ambient EDM track by David Miles Huber, with constant shaker sound in the height layer, taken from Auro-3D demo disc vol. 2.
- Everest: a soundtrack of movie “Everest”, with high frequency icy wind sound in the height layer, taken from Dolby Atmos demo disc 2017.
- Orchestra: an orchestral recording of the new year concert by Vienna Philharmonic, taken from Auro-3D demo disc Vol. 2.

Denon X4100W AVR was used for the decoding of the original content. The height channel signals of the original samples were processed using the VHAP VST plugin [3] on the Reaper digital audio workstation. As shown in Fig. 2, the plugin allows the user to locate multiple audio object (the height channel signals of the original samples in the present study) at different azimuth and elevation angles on a top-down view of the upper hemisphere.

Two types of VHAP stimuli were created for this experiment: 4-channel VHAP and 3-channel VHAP.

The original 4-channel VHAP method requires four loudspeakers in a cross layout as mentioned earlier. The four height channel signals were panned to the corresponding loudspeaker positions using the plugin: $\pm 45^\circ$ azimuth/ $\pm 45^\circ$ elevation for Front Left height (FLh) and Front Right height (FRh), and $\pm 135^\circ$ azimuth/ $\pm 45^\circ$ elevation for Rear Left height (RLh) and Rear Right height (RRh), as can be seen in Fig. 2 (a).

For the 3-channel VHAP, it was hypothesised that VHAP might still be effective if the Front Centre (FC), Rear Left (RL) and Rear Right (RR) loudspeakers of the conventional 5.1 layout were used, based on the localisation listening test results obtained in [1, 2]. Therefore, in this experiment, the rear loudspeakers placed at $\pm 110^\circ$ were used instead of the originally proposed side loudspeakers at $\pm 90^\circ$. Since the back-centre channel of the original VHAP is omitted, the phantom sources could not be panned to the rear half of the hemisphere. and instead the rear height signals are panned to $\pm 90^\circ$ azimuth/ $\pm 45^\circ$ elevation (Fig. 2 (b)).

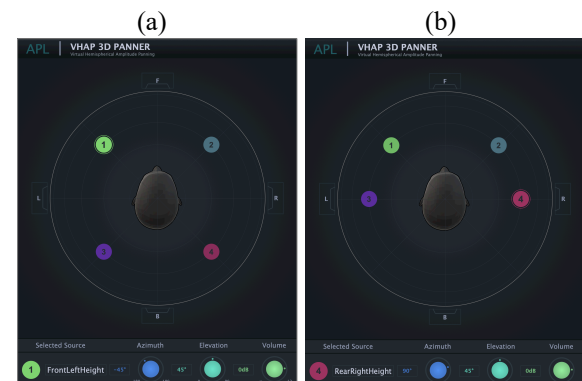


Fig. 2. VHAP VST plugin settings used for creating virtual height loudspeakers: (a) 4-channel VHAP (original proposal), (b) 3-channel VHAP (without the rear centre channel). Note that for the 3-channel version, the left and right loudspeakers were placed at $\pm 110^\circ$ instead of $\pm 90^\circ$, with the conventional 5.1 setup in mind.

The 3-channel version might compromise the perceived positions of the rear height images. However, previous research [1] suggests that the

perceived position of a VHAP image panned to a highly elevated position at $\pm 90^\circ$ azimuth tends to be biased towards the back slightly. Furthermore, with practical Dolby Atmos and Auro-3D content in film and pop music genres, the height signals are typically amplitude panned between the height loudspeakers, which means that each height signal would not be localised discretely. Therefore, it was expected that the 3-channel VHAP as well as the 4-channel one would render height images effectively compared to using real height loudspeakers.

The listening tests were conducted in an ITU-R BS.1116-compliant listening room at the Applied Psychoacoustics Laboratory of the University of Huddersfield (6.2m x 5.6m x 3.8m; RT=0.25s; NR 12). Prior to the listening test, the sound pressure levels (SPLs) of all stimulus were measured and calibrated at the listening position, using the Cassella CEL-63x sound level meter. The stimuli were presented via the Merging Technologies Horus interface. The loudspeakers used were five Genelec 8331As for the ear-height layer and four Genelec 8040As for the height layer. In the first stage of level calibration, the LAeq of the height layer of each VHAP stimulus was measured and adjusted to match that of the height layer of the original content within the tolerance range of ± 0.3 dBA tolerance. The level-matched VHAP signals were mixed with the original main layer signals, i.e., the VHAP signals were the subset of the 5.1. Finally, the playback levels (LAeq) of the stimuli were calibrated as follows: Airplane 75.4 dB, Bird 72.3 dB, EDM 75.2 dB, Everest 75.2 dB and Orchestra 73.7 dB. The levels were considered by the authors and subjects to be practical for the types of content.

2.2 Listening Test

Three sets of bipolar multiple comparison tests were conducted to evaluate the performance of VHAP in terms of the sense of height, the sense of width and overall quality of experience (QoE). In each trial, the 5.1 with 4-channel VHAP, 5.1 with 3-channel VHAP and hidden reference were rated against the reference (the original 5.1.4 content), using a scale ranging between -30 and 30. Semantic labels were provided on the scale: no difference to the reference at 0, slightly greater (0 to 10)/slightly lesser (-10 to

0), greater (10 to 20)/lesser (-20 to -10) and much greater (20 to 30)/much lesser (-30 to -20). The test interface was generated using the HULTI-GEN software [4]. The orders of the tests, trials and stimuli within each trial were randomised for each subject.

Ten subjects participated in the listening tests. They comprised post-graduate researchers and final year undergraduate students at the music technology courses of the University of Huddersfield. They all reported to have normal hearing. Six of them had many previous experiences in spatial audio listening test.

The subject's listening position was aligned to be exactly in the centre of the listening area using laser pointers. Their head position was not fixed, but they were asked to face forwards and stay still while evaluating the stimuli. This was visually monitored by the second author during the test.

3 Results

Shapiro-Wilks test suggested that the data obtained from the listening tests were not normally distributed for most conditions. Therefore, the medians and associated notch edges (i.e. non-parametric 95% confidence intervals) of the data are plotted in Fig. 3. For the same reason, Wilcoxon signed rank test was used for testing the significance of difference between each condition.

Overall, the differences of both VHAP methods to the original 5.1.4 content appear to be minimal for all three attributes tested. Wilcoxon tests suggest that there were only a few VHAP conditions that had a statistically significant degradation from the original, albeit only slightly (i.e., medians between 0 and -10). Those conditions were VHAP_4ch with Orchestra ($p=0.024$) for Sense of Height, VHAP_3ch with Aeroplane for Sense of Width ($p=0.011$), VHAP_4ch with Birds ($p=0.004$), VHAP_3ch ($p=0.017$) and VHAP_4ch ($p=0.000$) with Orchestra for QoE.

Although not significant, the medians of some VHAP_3ch conditions were slightly higher than the

original (e.g. Aeroplane, Birds and Everest for Sense of Height, and Aeroplane for QoE).

Furthermore, there was no significant difference found between the 3-channel and 4-channel VHAP methods ($p>0.05$).

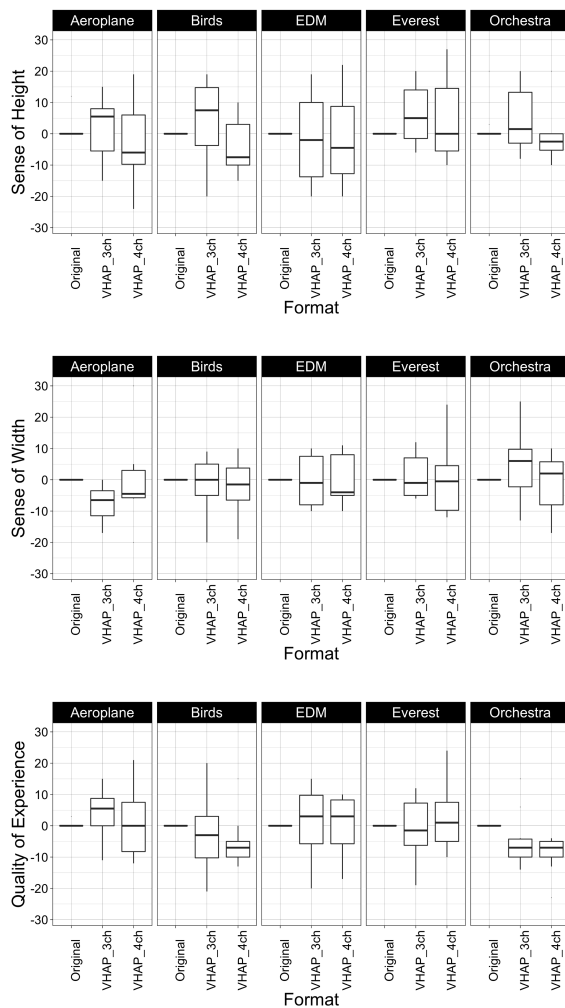


Fig. 3. Results of the listening tests: medians and associated notch edges (non-parametric 95% confidence intervals).

4 Discussion

The results support that VHAP can be an effective tool to create virtual height loudspeakers using only three or four ear-height loudspeakers. There was no significant degradation in QoE as well as in the perceived auditory height and width, apart from a few conditions that had only a slight degradation. The number of content samples used in this study was somewhat limited, but it is considered that the samples had some of the typical characteristics of height channel signals used in practical Dolby Atmos and Auro-3D content of different genres.

The 3-channel VHAP, which uses the front centre (0°), rear left (110°) and rear right loudspeakers (-110°) at the ear height, had no significant difference to or a slightly higher rating than the original 4-channel VHAP. This implies that VHAP does not need the rear centre loudspeaker in practice, and can be used directly with the conventional 5.1 loudspeaker configuration.

Therefore, it is considered that the 3-channel VHAP would be a useful tool for AV receivers that are equipped with Dolby Atmos and Auro-3D decoders. Home cinema users who already own a 5.1 loudspeaker system but have a practical constraint in installing additional height loudspeakers would benefit from such a feature. Another possible application is to add height information in 5.1 mixing stage using the VHAP plugin. Furthermore, in [3] it was shown that the binaural rendering of VHAP also produced effective elevation perception and externalisation. Therefore, it could be used for a more efficient binaural rendering of Dolby Atmos or Auro-3D content.

In addition, the current results seem to suggest that the overall quality of experience might have been influenced more by the height perception than the width perception. For example, the 4-channel VHAP had a slightly but significantly lower QoE than the original for Birds and Orchestra. It also had a slightly lower Sense of Height for the same sources, whereas there was no significant difference in Sense of Width. Another evidence might be that the QoE of the 3-channel VHAP was rated slightly higher

than the original for Aeroplane, when its Sense of Height was rated slightly higher but Sense of Width was rated significantly lower. Another factor that affected the QoE results may be the difference in tonal characteristics. VHAP uses ear-height loudspeakers, which produce different ear-input frequency spectra (i.e., HRTF) to the elevated loudspeakers used for the original content.

5 Conclusion

VHAP was used to render virtual height loudspeaker images for various Dolby Atmos 5.1.4 and Auro-3D 9.1 content materials. Listening tests were conducted to rate the 4-channel and 3-channel VHAP methods against the original content. From the results, it is concluded that the VHAP-rendered 5.1 reproduction can provide the listener with a similar or the same quality of experience as well as height and width impression compared with the reproduction with real height loudspeakers.

References

- [1] H. Lee, D. Johnson and M. Mironovs, "Virtual Hemispherical Amplitude Panning (VHAP): A Method for 3D Panning without Elevated Loudspeakers," presented in the 144th AES Convention, Preprint 9965 (2018)
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- [4] C. Gribben and H. Lee, "Towards the development of a universal listening test interface generator in Max", presented at the 138th Annual Audio Engineering Society AES Convention, Warsaw, Poland, 2015.